Tri Nguyen

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Education

The University of Texas at San Antonio May 2028
Bachelor of Science in Computer Science, Specialization in Data Science & ML *GPA: 3.93/4.0*

Experience

Better Future Insitute September 2026 – Present
Software Engineer Intern *San Antonio, TX*

- Redeveloped the front-end and refactored back-end code for the BUFFI system in a fast-paced environment, improving code quality and maintainability for company-facing clients
- Built data pipelines to gather and process client data, and implemented new features based on client requirements to support evolving business needs

Vorpex May 2026 - August 2026
Software Developer Itern *San Antonio, TX*

- Built and deployed a high-performance marketing website for Vorpex using Next.js 14 App Router and Tailwind CSS
- Integrated Cal.com booking, Resend transactional email, and Plausible Analytics into a static Next.js site, implementing custom API routes with rate limiting and honeypot spam protection

CooledTured LLC April 2026 - August 2026
Software Engineer Intern *Bensenville, IL*

- Building authentication systems, cloud deployment infrastructure, and data management pipelines using AWS, Kubernetes, and Docker
- Developing full-stack features for player registration, tournament organization, match reporting, and store management systems

Association Computing Machinery October 2025 – Present
Secretary *San Antonio, TX*

- Coordinated event logistics including room bookings, inventory management, and organizational planning for ACM events
- Collaborated closely with the President and Vice President to ensure smooth operations and execution of ACM initiatives and events serving 2000+ student members

Projects

Vision | *Python, React, JavaScript, MongoDB atlas, HasData API, Fast API, Leaflet* March 2026 – March 2026

- Selected as 1 of 15 teams in the USA for Redbull Basement Competition (0.2%)
- Integrated a Retrieval-Augmented Generation (RAG) pipeline using Google Gemini Pro to generate structural code compliance explanations from a curated engineering knowledge base.
- Built a deterministic structural analysis engine implementing AISC 360-16 and ASCE 7-22 load combinations to evaluate beam capacity, deflection, and code compliance.

LarpCity | *Python, React, TypeScript, Tigerdata PixiJS, RAG, Gemini, Vultr* February 2026 – February 2026

- Built LarpCity, a HackRice 2026 winning project, using TypeScript, Vite, and PixiJS to simulate decades of real-world financial decisions.
- Engineered a seeded financial simulation engine modeling debt, credit, investments, taxes, income, expenses, and major life events.
- Integrated real-world financial data and AI coaching to analyze decisions and provide personalized feedback on long-term financial outcomes.

Vending Machine Robot | *Python, Computer Vision, Pytorch, OpenVino* August 2025 – May 2026

- Developed real-time human detection and tracking systems using OpenVINO for autonomous robot interaction
- Engineered robot movement and hardware control logic to support autonomous navigation and operational reliability

- Implemented automated pathfinding and navigation using computer vision techniques for obstacle-aware robot movement

Radar | *Python, Computer Vision, Swift, OpenVino* September 2026 – September 2026

- Built a Wi-Fi sensing module to detect intruder movement, VTHacks 2026 winning project, and designed the iPhone/Apple Watch UI (SwiftUI) for real-time safety alerts
- Developed the Master Agent orchestrating a multi-agent AI backend, coordinating automated 911 calls with live incident data
- Deployed the multi-agent backend and web dashboard on Vultr for real-time monitoring and post-incident review

Mira | *Python, Computer Vision, Swift, TigerData* October 2026 – October 2026

- Won 1st Overall and MLH Best Use of Tiger Data at CodeRED Orion with Mira, an at-home PT coach that turns a webcam into a live 3D patient model.
- Built the backend (FastAPI, WebSockets, TimescaleDB) that streams 30 Hz pose data, with 41× faster queries and 2.74× lossless compression.
- Designed four agents (three using Gemini) for coaching, fall alerts, insights and therapist Q&A, grounded in real patient data.

Stochastic | *Python, FastAPI, Factory Droid SDK, Figma, Claude, TigerData* October 2026 – October 2026

- Built Stochastic at Cal Hacks 13.0 (Hyphenate track): four AI agents that merge an acquired company's books, ask a human only about the lines that matter, and never do the math themselves.
- Owned the finance tool layer (consolidation, EBITDA, covenant debt, leverage, limit test) in Python, verified against a real credit agreement, with a 435-test pytest suite.
- Built the FastAPI routes for CSV deal upload, SSE stage streaming, controller edits, the compliance certificate, and final workbook export, backed by Postgres/TimescaleDB on Tiger Cloud.

Technical Skills

Programming: Java, Python, C, Git, Docker, Kubernetes, SQL, TypeScript
AI/ML & Data Science: Pandas, NumPy, OpenVino, AI/ML (RAG, LLMS, PyTorch, Scikit-learn, data preprocessing)